

# PPT Lane — Handoff

**Version 1 · valid as of 2026-09-05.** For the 2 people building the deck. **Owner of this file:** Aamir. **Regenerate the PDF** with `python -m scripts.md2pdf docs/project/PPT_LANE_HANDOFF.md`.

Slots marked [ ] PENDING are content that does not exist yet. Each says **who** closes it and **when**. Build the slide now with the slot visible — an empty box with a label is a task; an empty box without one is a hole you find at 2 a.m.

## 0. What you are presenting, in judge language

Radar receivers can only listen to one narrow slice of the spectrum at a time, but they have to watch all of it. Today they sweep on a fixed schedule set before the mission — so they spend the same time on an empty band as on the one where a new threat just appeared. We built a simulator of that problem grounded in a real radar dataset, proved the simulator is faithful, measured seven classical scheduling strategies on it, and trained a scheduler that learns where to listen from nothing but its own hits and misses.

That is the pitch. Everything else on every slide is evidence for one of those five clauses.

**The problem statement is DRDO's, ID 26055, "Smart Scan strategy for Electronic Warfare."** Its expected solution, verbatim: *"Machine learning based Electronic Support receiver scheduler software."* Quote that line on the deck — it proves you read the ask.

## 1. Before anything else — two blocking checks

**1. Get the official SIH 2026 PPT template and follow it exactly.** Slide count, order and format are scored, and the playbook's submission checklist lists *"follows the official SIH PPT template exactly"* as a pass/fail item. **I have not verified what the 2026 template contains** — it is not in this repository, and nothing here should be treated as a substitute for it. If the official template conflicts with the structure in §3 below, **the official template wins** and §3 becomes a content checklist to redistribute across its slides.

[ ] PENDING — **official template downloaded and its slide list written into §3.** Owner: PPT lane. **Day 1, before any slide is designed.**

**2. There are two artefacts, not one, and they have different jobs.**

|               | Idea-submission deck                   | Finale / demo deck                           |
|---------------|----------------------------------------|----------------------------------------------|
| audience      | screening, seconds per slide           | live judges, questions                       |
| governed by   | the official template, strictly        | you                                          |
| optimised for | <i>is this real and is it feasible</i> | <i>do we believe the numbers</i>             |
| our strength  | the measured baseline table            | the live demo + the "what didn't work" slide |

Build the idea deck first. The finale deck reuses its slides plus depth.

## 2. Work split for two

| Role                    | Owns                                                                                                        |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| P1 — Narrative & design | the story order, the one-sentence claim, slide layout, visual consistency, the script                       |
| P2 — Evidence & figures | every number and figure on every slide, traced to a file and a section; the demo video; the reference slide |

**The rule that makes this work: P2 has veto over any number P1 wants to say.** Every figure on the deck must come from EVALUATION.md, DECISIONS.md, or a runs/ artefact — never from memory, never from an earlier draft of the deck. This project has already withdrawn four published figures that were repeated from documents instead of re-measured (D33, D41 and two in the 2026-09-03 audit). **Do not become the fifth.**

Both of you rehearse. The playbook's own readiness test: *any* member, not just the presenter, can walk a stranger through problem → solution → demo in under two minutes.

## 3. Slide-by-slide

Structure follows the deck order in docs/reference/PPT/SIH\_2026\_Playbook\_TechDoodles\_Final.pdf §08 (p.11), reordered where our project's strength differs from a generic one — **our evidence is unusually strong, so results move earlier than the playbook suggests.**

### Slide 1 — Title & team

PS 26055, "Smart Scan Strategy for Electronic Warfare", DRDO. Team name, institution, member names and roles.

[ ] **PENDING** — team name, member list. Owner: Aamir.

### Slide 2 — The problem, made physical

Not a paragraph of text. **One diagram:** 36 bands stacked vertically, time across, emitters lighting up, and a single receiver cursor that can only be in one place at a time. The instant someone sees that picture they understand the problem.

Say: the receiver's instantaneous bandwidth is *an order of magnitude smaller* than the spectrum it must cover (PS, Background). Today's strategies are **open loop** — fixed before the mission — and by the PS's own words "*may lose time to nonthreatening emitters by not giving time to new or threatening ones.*" **Quote the PS here.**

Source: SIH26055\_PROBLEM\_STATEMENT.md. Figure: build from runs/baselines/timeline\_config\_2.png, which is exactly this picture with real data in it.

### Slide 3 — Proposed solution, in one sentence

One sentence, large. Then three bullets: **a validated simulator, a measured baseline ladder, a learned scheduler.** Nothing else on this slide.

[ ] **PENDING** — the one sentence, and the tagline. Owner: Aamir + the scheduler brainstorm. Closed by end of RL Day 2. See `RL_LANE_HANDOFF.md` §9 for the four candidates and the method.

## Slide 4 — Innovation & uniqueness

This is the slide that decides whether you stand out, and it is currently the emptiest.

The candidates, with the number that backs each — pick **one headline and one supporting**, do not list all four:

| Angle                                                                                               | The number behind it                                                                                                  | Status                               |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>The operator's dial</b> — the two objectives provably conflict, so we ship a front, not a policy | D28: censored intercept time needs $Y=1$ , interception ratio does not — no single reward serves both                 | needs a preference-conditioned agent |
| <b>Learning to see what the receiver can't measure</b>                                              | truth-fed camper 0.568 vs observation-fed 0.209 — a <b>0.359 gap</b> that is the price of invisible information (D46) | measured ■                           |
| <b>We can prove our simulator is right</b>                                                          | 4 gates, criteria fixed <i>before</i> the run and asserted by a test (D39); D42 publishes the gates' own blind spots  | measured ■                           |
| <b>What didn't work</b>                                                                             | Apfeld's published period estimation <i>hurt</i> : 1.9× ratio for 3.4× the intercept time (D45)                       | measured ■                           |

[ ] **PENDING** — the chosen angle. Owner: Aamir + brainstorm. **Blocks slides 3, 4 and 12.**

## Slide 5 — How it works

The loop, as a diagram, four boxes: **observe** (own scan history only) → **decide** (which band) → **listen** (one dwell, 50 or 100 ms) → **learn** (hit or miss). Then the line that makes it credible to anyone technical:

The agent sees **only what a real receiver would see** — its own hits and misses. Ground truth exists in the simulator, and it is used to *score* the agent, never to *inform* it.

That asymmetry (D29) is a real engineering discipline and it is worth 20 seconds of airtime.

## Slide 6 — Technical approach & architecture

The four-layer stack, one line each: **L0 scenario** (emitters drawn from real recordings) → **L1 truth grid** (36 bands × 600 slots, who is transmitting when) → **L2 receiver** (a detector with a frozen threshold and a measured ROC) → **L3 agent interface** (a standard Gymnasium environment).

Stack: Python, NumPy, HDF5, Gymnasium, PyTorch/Stable-Baselines3, Matplotlib. **229 automated tests**. Say the test count — almost no SIH deck can.

Source: `ENVIRONMENT_SPEC.md`, `rfenv/` module docstrings.

## Slide 7 — The data is real, and here is how we used it

The Turing Synthetic Radar Dataset: **184 HDF5 files, 2.2 GB, 47 train + 45 held-out scenario pairs**. The held-out 45 are touched **once, at the end** (D8).

Two facts that show you actually opened the files rather than reading a dataset card: - **19% of transmitters in the metadata are never detectable at all** — scoring a scheduler for missing those would be meaningless, so our coverage denominator excludes them (D27). - **A scan recording contains only the pulses the dataset's own sweeping receiver was tuned to**. Benchmark on one and any sweeping scheduler scores 0.9999 — we caught that and excluded scan replays from every comparison (D36).

The second one is a *strong* slide. It says: we understood our data well enough to find the trap inside it.

## Slide 8 — Results: the baseline ladder

The table from `EVALUATION.md` §5, plus `runs/baselines/pareto.png`. State beneath it: **1,539 episodes, 3 seeds,  $\gamma = -111$  dB, 2026-09-04**.

Three rules, non-negotiable: 1. **Never show interception ratio alone**. Ratio, censored intercept time and coverage together, always (D14). A camper scores 0.209 and looks great until 9.67 s and 0.497 coverage appear. 2. **Label the two oracle rows as reference lines** wherever they appear. They read the truth grid. Read as competitors they invert the meaning of the whole table. 3. **The bar is rung 5, not round-robin**. Say so. Claiming victory over a weak baseline is the fastest way to lose a technical judge.

[ ] **PENDING** — the **rung 7 row**. Owner: RL lane, end of Day 3. Build the slide with the row present and empty; do not restructure it later.

## Slide 9 — Results: the agent

Before/after: rung 5 vs rung 7. `pareto.png` with rung 7 marked, and `discovery_config_921.png` (distinct emitters found against time — the most intuitive figure we have).

[ ] **PENDING** — all numbers. Owner: RL lane, end of Day 3.

## Slide 10 — Feasibility, and why you should believe the numbers

Feasibility here is not "can it be built" — it is built. It is "**is the simulator faithful?**", and that is our strongest and least-copyable slide:

- **Gate 1** — build truth from *stare* recordings only, replay the *scan* schedule, predict against the real scan recordings. Data never used in construction. **Accuracy 0.859, MCC 0.685** over 23,594 dwells.
- **Gate 2** — pipeline self-consistency, per band. **0.000 pp** deviation.
- **Gate 3** — a controlled periodic case against published closed-form theory (Köksal). **Exact**.
- **Gate 4** — 12 structural assertions on the two extreme scenarios. **12/12**.

Then the line that separates us from every team that shows a validation slide:

**Every pass criterion was written into the code before the first run, and a test asserts it against the record** (D39). And we publish what the gates *cannot* detect (D42) — including that no gate covers the receiver's band half-width.

**Also state the known limitation, on the slide:** band 0 is 59% occupied in the recordings and 0% predicted, because the stare recordings cannot see below 500 MHz (D10). Naming your own limitation before a judge finds it converts a weakness into evidence of rigour.

## Slide 11 — Impact & scalability

Who benefits, and the measurable outcome: a receiver that finds threats sooner, at no hardware cost — it is a **scheduling policy**, deployable as software on the receiver that already exists. Scalability: nothing in the method is specific to 36 bands or to this dataset.

[ ] **PENDING** — impact framing. Owner: P1. **Keep it honest.** Differentiation-doc §28: do not claim superiority over fielded or classified military systems. We have a simulator result, and that is a strong, defensible claim on its own.

Slide 12 — What we tried that did not work

One slide. Four bullets, from DECISIONS.md:

- **D45** — we implemented a published adaptive method; its period-estimation half made things worse here, and lost to its own ablation.
- **D44** — that paper's Algorithm 1 contradicts its own prose. We implement the prose and say so.
- **D43** — two of our own baselines turned out to be the same policy. We found it and split them.
- **D41 / D33** — we withdrew our own published figures when their measurement convention turned out to be unrecorded.

Most decks cannot fill this slide. It is disproportionately persuasive — it is the difference between a team that got a number and a team that understands its number.

Slide 13 — Team & roles

Why this team can execute *this* PS.

[ ] **PENDING** — owner: Aamir.

Slide 14 — References

The PS, the Turing dataset, the two scheduling papers (optimumsearch.pdf — Köksal; paperSSPD (1).pdf — Apfeld et al.), Gymnasium. Cite the papers properly — we adapted one and contradicted the other, and both are named honestly.

4. Figures that already exist

All in runs/baselines/, regenerated by one command:

```
python -m rfenv.compare --seeds 3 --sampled 10 --figures --out runs/baselines
```

| File                     | What it shows                                                                                                   | Best slide |
|--------------------------|-----------------------------------------------------------------------------------------------------------------|------------|
| pareto.png               | interception ratio vs censored intercept time, one marker per rung, marker area = coverage                      | 8, 9       |
| timeline_config_2.png    | band vs time with occupancy, the tuning path, hits, and a per-emitter strip that turns solid at first intercept | 2, 5       |
| discovery_config_921.png | distinct emitters found against time, one line per rung, with the ceiling                                       | 9          |

| File                                            | What it shows                                                                   | Best slide |
|-------------------------------------------------|---------------------------------------------------------------------------------|------------|
| <code>timeline_config_81.png</code>             | the 1-emitter extreme — good for explaining the mechanism slowly                | 5          |
| the waterfall ( <code>render.waterfall</code> ) | the 36×600 grid as a frequency-vs-time heatmap — the standard ESM operator view | 2, 7       |

`runs/` is **gitignored** — these are rebuilt, not committed. Regenerate before exporting slides so the figures match the numbers beside them.

**Design note:** they are Matplotlib defaults. They are honest but not presentation-grade. Restyle for the deck (larger fonts, fewer gridlines, one accent colour), **but never redraw a figure by hand or in a spreadsheet** — a hand-drawn chart can disagree with the table beside it, which is precisely why `render.py` draws all of them from the same artefacts.

[ ] **PENDING** — a diagram of the 4-layer architecture for slide 6. Owner: P1. Not auto-generated; draw it.

## 5. Traps

1. **Never quote a number from an earlier draft of the deck.** Re-open the source. Four published figures in this project have already been withdrawn for exactly this.
2. **State the operating point** ( $\gamma = -111$  dB,  $P_{fa} = 1.35e-3$ ) beside any scheduler table, and name which population  $P_d$  was averaged over — it is not data-independent (D33, D46).
3. **Do not say "accuracy"** as a headline. With sparse occupancy, "predict nothing" scores well.
4. **Do not call the oracle rows baselines.**
5. **RL\_LANE\_HANDOFF.pdf and .html in this folder are stale.** Ignore them; the `.md` files are the source.
6. **Nothing from `../SIHProto`.** Not a number, not a diagram, not a slide.
7. **Don't oversell.** We have a validated simulator and a measured comparison. That is genuinely more than most entries have, and it survives questioning — which a bigger claim will not.

## 6. Timeline

| Day | P1 (narrative)                                                      | P2 (evidence)                                                                        |
|-----|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1   | official template downloaded; slide skeleton with every [ ] visible | pull every existing number into one sheet with its source; regenerate figures        |
| 2   | slides 1, 2, 5, 6, 7 fully drafted                                  | slides 8, 10, 12 built from <code>EVALUATION.md</code> and <code>DECISIONS.md</code> |
| 3   | slides 3, 4, 11 once the edge is chosen                             | rung 7 numbers land → slides 8, 9                                                    |

| Day | P1 (narrative)                                    | P2 (evidence)                                                                                    |
|-----|---------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 4   | full pass for consistency and one visual language | <b>record the backup demo video locally;</b><br>verify every number against its source once more |
| 5   | two timed rehearsals, both members presenting     | held-out results land → final number pass                                                        |
| 6   | buffer                                            | buffer                                                                                           |

## 7. Submission checklist

From the playbook (p.12), kept only where it applies to us:

- ☐ Follows the official SIH PPT template exactly — slide count, order, format
- ☐ Every slide answers one clear question; no filler slides
- ☐ Core end-to-end flow works, even if unpolished
- ☐ **Backup demo video recorded locally** — the playbook's most-cited live failure
- ☐ Rehearsed as a team at least twice, against a timer
- ☐ Every member can explain the PS without notes
- ☐ Every number on every slide traced to a file and a section by P2
- ☐ Every reference line labelled as a reference line
- ☐ Named beneficiary and measurable outcome present
- ☐ No claim of superiority over fielded or classified systems

## 8. Where the content lives

| Need                                        | Go to                                                                                                |
|---------------------------------------------|------------------------------------------------------------------------------------------------------|
| the ask, verbatim                           | <code>docs/project/SIH26055_PROBLEM_STATEMENT.md</code>                                              |
| every metric, baseline, gate figure         | <code>docs/project/EVALUATION.md</code> §4, §5, §6                                                   |
| "what we tried and what changed our mind"   | <code>docs/project/DECISIONS.md</code> — D33, D36, D41, D43, D44, D45, D46                           |
| the architecture, in words                  | <code>docs/project/PROJECT_ARCHITECTURE.md</code> ,<br><code>docs/project/ENVIRONMENT_SPEC.md</code> |
| what the RL lane is doing and when it lands | <code>docs/project/RL_LANE_HANDOFF.md</code>                                                         |

| Need                           | Go to                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------|
| a claim inside a reference PDF | <code>.venv/bin/python -m docsearch "your question" -k 8</code> — returns <code>file.pdf:p.7</code> |
| past winners' decks            | <code>docs/reference/PPT/</code> — 7 of the 10 are image-only, open them visually                   |